In Situ Temperature Measurement Test Report

For

American Lighting Industry Corp

(Brand Name: ASTRE LIGHTS)

7 Scouting Blvd,medford, NY 11763

Outdoor Pole/Arm-Mounted Area and Roadway Luminaires

Model name(s):
ALI-SBX-200/240/300W-5000/5700/6500K-H3-S1[Blank;(PIR);(PTC);(MS);(PIP);(MSP);(PR);(PSC);(BT)]

Remark: The "[Blank;(PIR);(PTC);(MS);(PIP);(MSP);(PR);(PSC);(BT)]" can be (PIR)=PIR sensor, (PTC)=Photocell; (MS)=Motion sensor; (PIP)=PIR sensor and Photocell, (MSP)=Motion sensor and Photocell; (PR)=PR Motion sensor; (PSC)=Photocell socket and shorting cap; (BT)=Bluetooth Motion sensor or blank=no sensor function.

Representative (Tested) Model:
ALI-SBX-200/240/300W-5000/5700/6500K-H3-S1(0%,5000K)
ALI-SBX-200/240/300W-5000/5700/6500K-H3-S1(100%,6500K)

Model differences: N/A

Test & Report By:

Ferrum Li

Engineer: Ferrum Li

Date: 2025-04-28

Review By:

Garman Mo

Manager: Garman Mo

Note: 1.The results contained in this report pertain only to the tested samples.

2.This report does not imply product certification, approval, or endorsement by A2LA, or any agency of the Federal Government.

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1 General

1.1 Product Information

Brand Name	ASTRE LIGHTS
Trade Mark	--
Type of Luminaire	Outdoor Pole/Arm-Mounted Area and Roadway Luminaires
Model Number	ALI-SBX-200/240/300W-5000/5700/6500K-H3-S1[Blank;(PIR);(PTC);(MS);(PIP);(MSP);(PR);(PSC);(BT)]
Rated Voltage / Frequency	120-277Vac, 50/60Hz
Nominal Power	200W/240W/300W (Power adjustable)
Rated Initial Lamp Lumens	--
Declared CCT	5000K,5700K,6500K(Color Tunable)
LED Manufacturer	Bridgelux Inc.
LED Package, Array or Module	BXEN-50E-13H-9D1 BXEN-65E-13H-9D1
Driver Model	SS-320NH-E260BH
Sample Number	JJDE250402-B1
Date of Receipt Samples	2025-04-11
Photo	

1.2 Standards or methods

The following standards are partly or totally used or referenced for test:

No.	Name
ANSI/UL 1598:2008	**Luminaires**

1.3 Equipment list

Equipment ID	Equipment Name	Last Calibration Date	Next Calibration Date
ST-R-S-384	Power Meter	2024-05-29	2025-05-28
ST-R-S-462	Temperature Tester	2024-05-29	2025-05-28

2 Test conducted and method

2.1 Ambient Condition

Test was conducted in an ambient temperature of 25±5 ℃. Ambient temperature variations above or below 25 ℃ was subtracted from or added to temperatures recorded at points on the luminaire.
The ambient temperature was measured by a thermocouple which was immersed in 15ml of mineral oil in a glass container.

2.2 Temperature Stabilization

Temperatures were measured after they have stabilized when the test has been running for a minimum of 7.5 hours, or the test has been running for a minimum of 3 hours and three successive reading taken at 15 minutes intervals are with 1 ℃ of another and are not rising.

2.3 Thermocouples

Type K thermocouple was used for temperature measurement. The thermocouple was 0.05mm2(30AWG), and complied with the requirements specified in ASTM MNL 12 and limits of error specified in NIST ITS 90 and ISA MC96.1.

2.4 Thermocouples contact

Thermocouples were in contact with the TMP LED location described in LM-80 test report. In order to gain the maximum temperature, if appropriate, more than one thermocouple were contact in these locations. For details information, please refer to clause 3.3 for the photo of thermocouple contact.

3 Test Results

Test date	2025-04-14	Test Ambient	25.1 ℃
Sample No.		LED Package Model	
JJDE250402-B1(0%,5000K)		BXEN-50E-13H-9D1	
LED driver of Each Lamp	Output voltage V	Measured LED working current (Max.) mA	
1	232.9	98.54	
Sample No.		LED Package Model	
JJDE250402-B1(100%,6500K)		BXEN-65E-13H-9D1	
LED driver of Each Lamp	Output voltage V	Measured LED working current (Max.) mA	
1	232.9	98.50	

3.1 Data:

(0%,5000K)

Input Vol.	120.0V	Input Current	2.391A	Input Wattage	285.0W	Temperature stabilization time:	500 min

No.	Temperature (°C)		No.	Temperature (°C)		No.	Temperature (°C)	
	Measured	Corrected at 25°C		Measured	Corrected at 25°C		Measured	Corrected at 25°C
1	61.1	61.0	2	60.7	60.6	3	60.9	60.8
The highest in-situ measured temperature LED is 61.0°C								

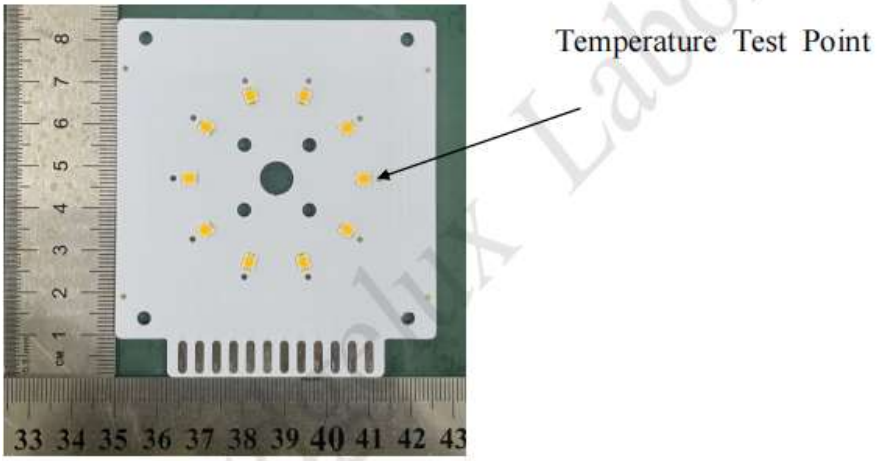
(100%,6500K)

Input Vol.	120.0V	Input Current	2.376A	Input Wattage	283.5W	Temperature stabilization time:	500 min

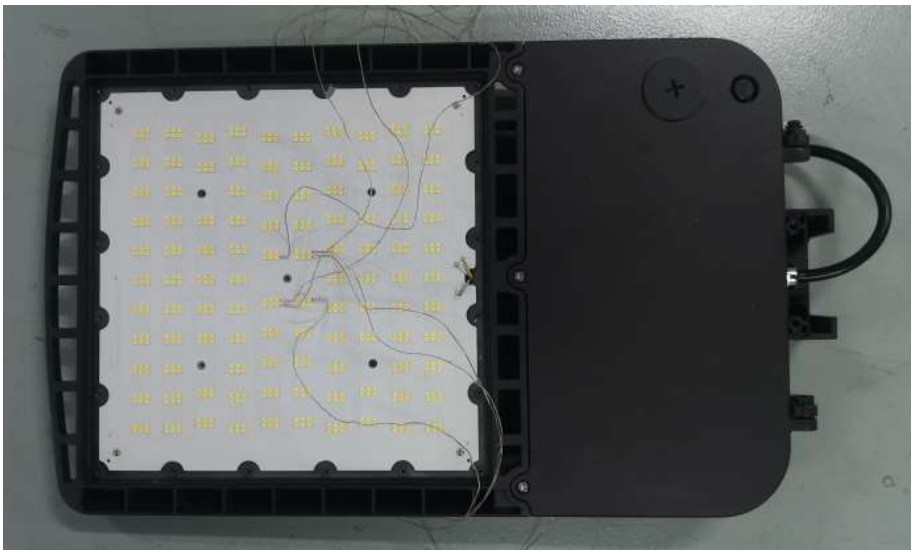
No.	Temperature (°C)		No.	Temperature (°C)		No.	Temperature (°C)	
	Measured	Corrected at 25°C		Measured	Corrected at 25°C		Measured	Corrected at 25°C
1	60.5	60.4	2	60.3	60.2	3	60.7	60.6
The highest in-situ measured temperature LED is 60.6°C								

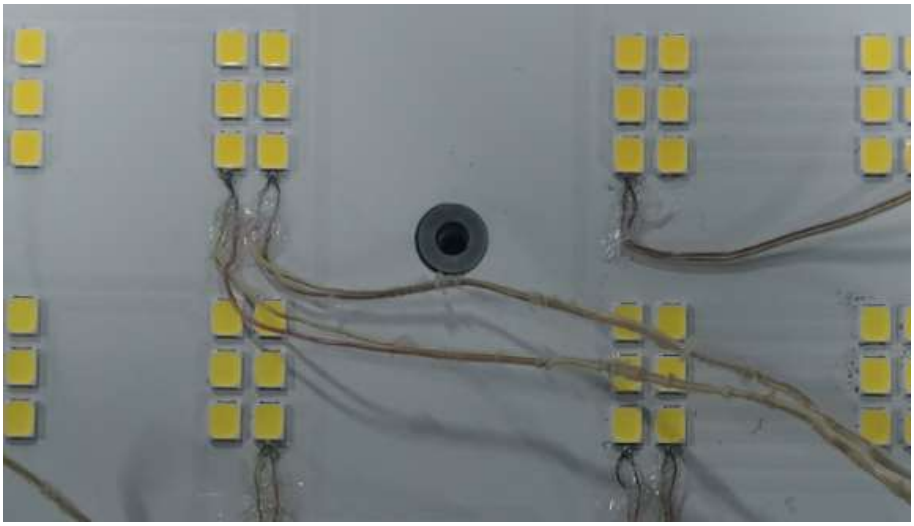
3.2 Test Photo:

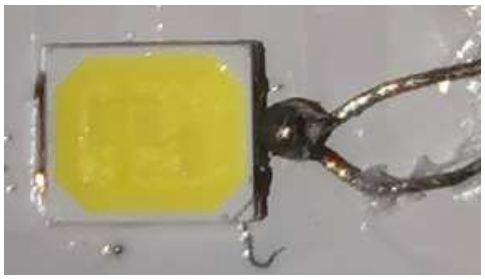
Ts Position:



Thermocouple Location on Temperature Measurement Point (TMP):







Time (t) at which to estimate lumen maintenance (hours):	50,000
Lumen maintenance at time (t) (%):	90.69%
Reported L70 (hours):	>102000

Time (t) at which to estimate lumen maintenance (hours):	36,000
Lumen maintenance at time (t) (%):	92.83%
Reported L90 (hours):	55,000

3.3 Test Data of LED Driver:

(0%,5000K)

Input Vol.	120.0V	Input Current	2.391A	Input Wattage	285.0W	Temperature stabilization time:	500 min

No	Measured TC Temperature (°C)		Temperature Limited of Life ⩾ 50000 hours
	Measured	Corrected at 25°C	
1	51.5	51.4	85

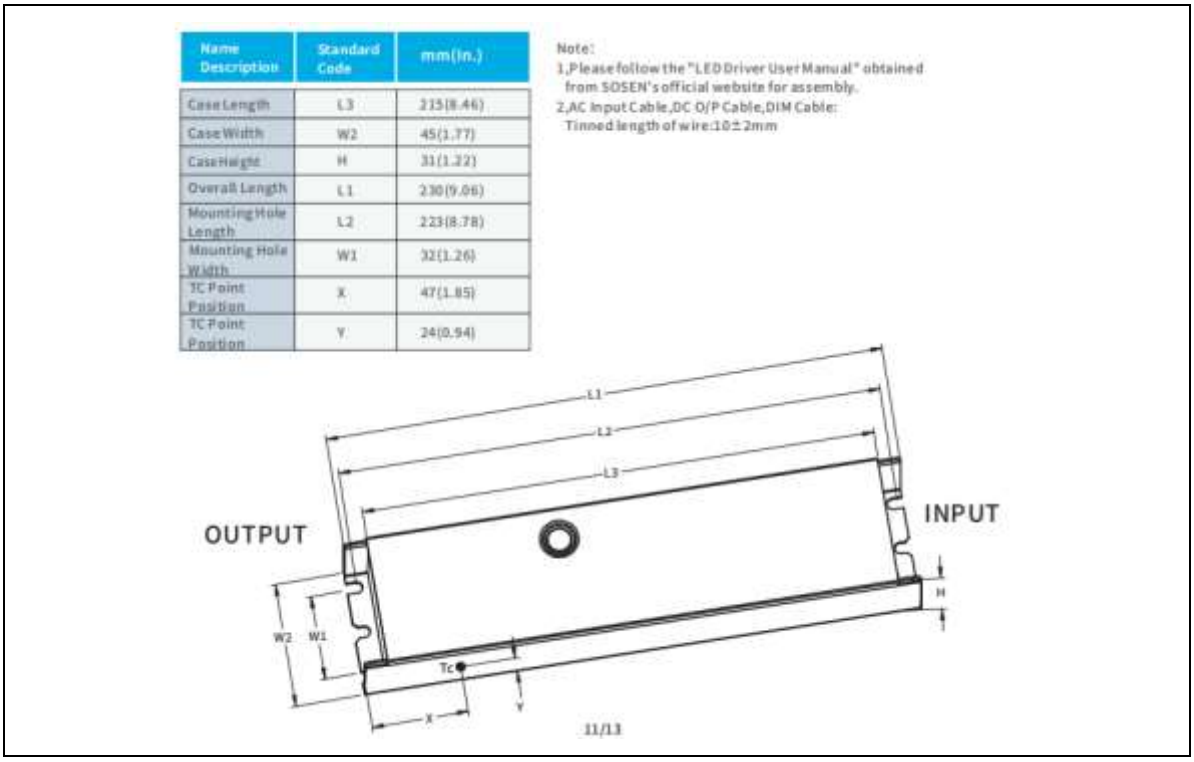
(100%,6500K)

Input Vol.	120.0V	Input Current	2.376A	Input Wattage	283.5W	Temperature stabilization time:	500 min

No	Measured TC Temperature (°C)		Temperature Limited of Life ⩾ 50000 hours
	Measured	Corrected at 25°C	
1	51.4	51.3	85

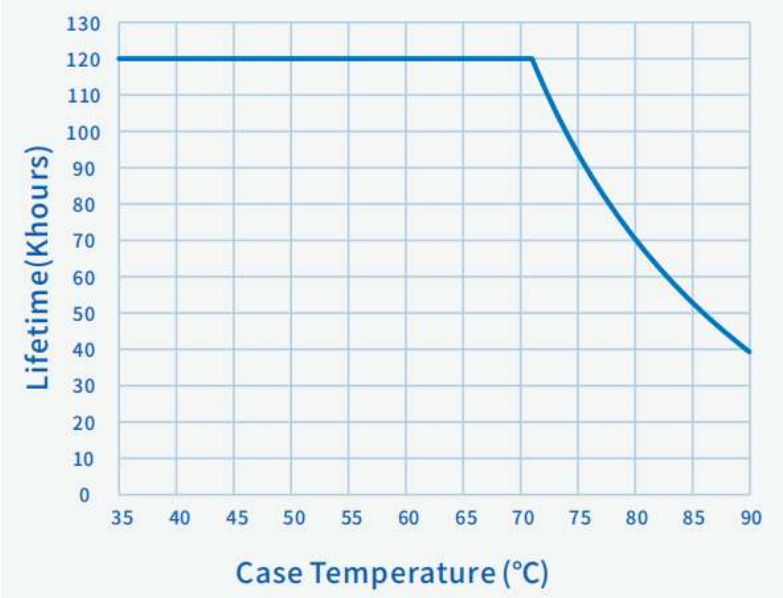
3.4 Test Photo:

Thermocouple Location on Temperature Measurement Point (TMP):



Name Description	Standard Code	mm(in.)
Case Length	L3	215(8.46)
Case Width	W2	45(1.77)
Case Height	H	31(1.22)
Overall Length	L1	230(9.06)
Mounting Hole Length	L2	223(8.78)
Mounting Hole Width	W1	32(1.26)
TC Point Position	X	47(1.85)
TC Point Position	Y	24(0.94)

Note:
1,Please follow the "LED Driver User Manual" obtained from SOSEN's official website for assembly.
2,AC Input Cable,DC O/P Cable,DIM Cable: Tinned length of wire:10±2mm



****** END OF THE TEST REPORT******